TAXGUARD

Proposal update guide

Purpose. The prototype has been renamed to **TaxGuard** and changed so it matches the technical proposal. This guide lists (1) what changed in the prototype, (2) the exact edits to make to the proposal, section by section, with ready-to-paste replacement text, and (3) the measured results that replace the proposal's "will evaluate" language. Every figure comes from the prototype's evaluation run on labelled synthetic data.

How to use it. Work through Part 2 in order. Text in a **blue box** is suggested wording you can paste and adapt. Items marked **[assumption]** are placeholders the economist must confirm.

Headline (synthetic data)	TaxGuard	Baseline: review the largest claims
Precision - flagged claims that are abusive	93%	38%
Recall - abusive claims caught	94%	39%
Legitimate refunds fast-tracked (complete records)	96%	66%
Abusive refund value flagged	M 237.2m	M 204.2m
Phantom-supplier invoices caught	571 / 571	-
Duplicate invoices caught (false alarms)	295 / 295 (0)	-
Flags with a plain-language explanation	100%	-

Dataset: 2,512 refund claims (883 abusive) from 417 claimant taxpayers, 2024-11 to 2026-10. The baseline flags the same number of claims (891) by picking the largest refunds (threshold M 180,082). Synthetic results cannot establish real-world accuracy - see Part 3.

Part 1 - What changed in the prototype

These changes close the gaps between the proposal and the working demo. The right-hand column is the proposal section each change now supports.

Change	What it does	Proposal section
Renamed to TaxGuard	App name, logo, page titles, API title, evaluation charts and docs now say TaxGuard.	Title, all
New rule R12 - related-party financing	Flags intra-group interest above 1% of turnover while the company reports losses in consecutive years. Group-structure rule: a mitigating factor cannot offset it.	§3.1, §4 (financing anomaly)
Missing / contradictory data labelled (D01-D03)	D01 financial statements missing, D02 VAT returns not filed, D03 returns disagree with financial statements by >30%. Shown separately, never scored as zero or high risk. Rules that could not run are listed. A claim with a data gap cannot be fast-tracked (safeguard L03) and the missing documents are requested.	§4 (missing data)
Per-flag review status	Officers mark each flag Reviewed, Not applicable or Information required, with a mandatory reason. Stored per claim, shown on the case and written to the audit log.	§4, §7.1
PDF / PNG / JPG upload with allowlist	Files are checked by content signature (not extension) and anything else is rejected. The first PDF page is read, and the original file is preserved unchanged. Fields are marked machine-extracted with confidence.	§3.2, §6
New test scenario: incomplete records	8 legitimate exporters with unfiled months, missing statements and a turnover mismatch, matching the proposal's "incomplete or contradictory records" scenario.	§4.1
Response-time and explainability measures	Evaluation now reports median / p95 response times and the share of flags with an explanation (100% of 6,313).	§8
Evaluation separates document requests	Claims sent to review only because records are missing are reported as document requests, not risk flags, and an overall figure including them is also given.	§8

Already in the prototype before this round (the proposal should mention these, see Part 2): supplier cross-match against suppliers' declared sales; duplicate and altered-copy detection; mitigating factors (exports, capital assets, clean history, seasonality); Isolation Forest anomaly model with plain-language drivers; related-party group graph; role-based access with masked TINs; mandatory decision reasons and override tracking; full audit log; admin-editable thresholds with re-scoring; taxpayer portal with a draft Sesotho toggle; client-side image compression for weak connections.

Part 2 - Edits to the proposal, section by section

Global fixes

- Use **TaxGuard** consistently (the cover uses TAXGUARD - fine for the logo, but use TaxGuard in body text).
- Fix the heading numbering: the second "3." heading and "3. 3.1" should read **3.1**; check the numbering after it.
- Check the Laudage Teles, Riedel & Strohmaier (2024) figures (5-8% corporate tax revenue increase) against the paper itself.
- Replace future-tense "will build / will evaluate" with what the prototype now does, and keep the honest caveats.

Executive summary - add two sentences

The summary is careful but doesn't say what's new. Add after the second paragraph:

> TaxGuard checks every supporting invoice against the supplier's own declared sales, catches duplicate and altered re-photographed invoices, and explains every flag in plain language with the evidence behind it. On 2,512 labelled synthetic claims it achieved 93% precision and 94% recall, against 38% and 39% for selecting the largest claims, while fast-tracking 96% of legitimate refunds with complete records.

§2.2 Problem statement - replace

The current text covers audit selection only. Taxpayer and business value is the largest judging criterion (25%), so the problem should be two-sided. Suggested replacement:

> Tax and financial information is spread across returns, schedules, invoices, financial statements and periods. Manual collation is slow, so refund verification creates a double cost. Legitimate claimants, especially exporters with zero-rated sales, wait for refunds that affect their cash flow, while abusive claims supported by unsupported, duplicate or phantom invoices can still be paid. Related-party structures add further complexity: a refund position, a loss or an intra-group charge does not on its own prove abuse. The Taxathon brief cites a preliminary estimate of around M1bn in annual tax forgone. RSL needs an analytical support layer that pays legitimate refunds faster, concentrates scarce audit capacity on the claims most likely to be abusive, explains why each case is flagged, and lets auditors trace each flag to the underlying evidence.

§3 Modules table - add or amend rows

Module	Function	User value
Supplier cross-match & duplicate detection (new row)	Checks each supporting invoice against the supplier's declared sales listing and VAT return, the VAT register, prior invoices (number, fields) and a perceptual image hash.	Catches phantom and reused invoices with evidence.
Related-party group explorer (new row)	Interactive graph of MNE group entities and intra-group flows; highlights paths from Lesotho entities to low-tax jurisdictions.	Makes group structures visible for Special Sectors.
Invoice Intake & Evidence Register (amend)	Accepts PDF, PNG, JPG (content-signature allowlist, 12 MB limit); OCR suggests fields with confidence; original preserved; manual route always available.	Implemented in the prototype.
Taxpayer intake (demo, amend §7.2)	Demonstration portal: submit claim, upload invoice photos, see status and document requests. Draft Sesotho labels.	Shown as a future option subject to RSL approval.

§4 Risk analytics and scoring framework - replace the indicator table

The proposal describes a weighted average of six indicators with bands at 30 and 60. The prototype instead adds up points from explainable rules, so use the text and table below. The weights are illustrative demo values, not RSL thresholds.

Each rule produces a severity between 0 and 1. Points = severity x the rule's maximum points, and the risk score (0-100) is the sum. Rules are grouped into three types. **Document evidence** and **group-structure** findings can never be offset. **Behaviour patterns** can be explained away by documented mitigating factors such as zero-rated exports or capital purchases. An Isolation Forest anomaly model adds up to 15 points and is always shown with its main drivers. Demo lanes: **Fast-track** below 15, **Officer review** 15-34, **Audit** 35 and above. Three safeguards override the score: a strong document finding is never fast-tracked (L01), a large capital-asset refund needs one supporting document (L02), and missing or contradictory records must be resolved first (L03).

Code	Indicator	Type	Max pts
R01	Persistent refund claimant (8+ of last 12 months)	Pattern	12
R02	Input VAT not supported by invoices (>10% gap)	Evidence	25
R03	Invoices from suppliers who did not declare the sale	Evidence	35
R04	Duplicate / altered / re-photographed invoices	Evidence	30
R05	Reverse-charge VAT missing on related-party imported services	Evidence	15
R06	Input/output VAT ratio far above sector peers	Pattern	10
R07	Refund spike versus own history	Pattern	10
R08	High-risk or newly registered suppliers	Pattern	12
R09	Related-party charges to low-tax entities > 5% of turnover	Group	20
R10	Persistent losses alongside growing turnover	Group	15
R11	Margin far below peers with heavy intra-group purchases	Group	20
R12	Related-party financing alongside losses (new)	Group	10
ANOM	Isolation Forest statistical anomaly	Pattern	15
M01-M04	Mitigations: exports, capital assets, clean history, seasonality	Offsets patterns	-
D01-D03	Data gaps: statements missing, returns not filed, records disagree	Not scored	-

Keep the proposal's good bullets on missing data, reviewer dispositions, versioning and human review - the prototype now implements all of them. Mapping to the proposal's bands: Low = Fast-track, Medium = Officer review, High = Audit.

§4.1 Scenarios: the six proposal scenarios map to prototype data - ordinary low-risk (normal taxpayers), recurring refund with valid invoices (exporters), high related-party expenses and losses (profit shifting), imported-service mismatch (reverse charge), incomplete or contradictory records (new), and a false positive with a commercial explanation (seasonal stock build-up and capital purchases). Also list the invoice-fraud scenarios: phantom suppliers, duplicates, unsupported input VAT.

§6 Security - split into "in the prototype" and "before a pilot"

As written, §6 reads as if the prototype already has MFA, encryption and malware scanning. It doesn't. Splitting the list is more honest and scores better on trust:

In the prototype now	Required before a pilot
Role-based access (auditor, analyst, Special Sectors, admin, taxpayer)	MFA via RSL identity management
TINs masked except for the assigned officer	TLS everywhere; encryption at rest for DB, backups, documents
Mandatory reason on every decision and flag review; overrides tracked	Malware scanning and OCR sandboxing for uploads
Audit log: logins, views, scans, scores, decisions, settings changes	Tamper-evident (append-only, hashed) audit log
Upload allowlist by file signature; 12 MB limit; originals preserved	Secure sessions, idle timeout, penetration test

In the prototype now	Required before a pilot
No automated rejection; ground-truth labels never loaded into the app	Retention schedule, DPIA, access matrix, incident response

§7.3 Usability - small update

- The prototype already has a responsive layout, plain-language reasons, client-side image compression and code-split pages (initial download ~90 KB gzipped).
- The taxpayer portal has an English / Sesotho toggle. The Sesotho labels are **draft** and must be checked by a native speaker; say so in the proposal.

§8 Performance evaluation - replace "proposed evaluation" with results

Dimension	Result on synthetic test data
Calculation correctness	9 automated tests pass (demo lanes, explanations, roles and masking, mandatory reasons, R12, data-gap labels, flag reviews, upload allowlist and PDF).
Flag detection (recall)	94% of abusive claims flagged (baseline 39%). The Audit lane alone: precision 100%, recall 50%.
False positives	Precision 93% (baseline 38%); false-positive rate 3.8%. 176 further claims were sent for missing records (0 abusive) - a document request, not a risk flag. Including them: precision 78%.
Explainability	100% of 6,313 triggered flags show code, rationale and evidence.
Responsiveness	Case view median 49 ms / p95 58 ms; queue median 64 ms; dashboard median 253 ms (single local process, SQLite, developer laptop).
Document intake	Phantom-supplier invoices 571/571; duplicates 295/295 with 0 false alarms; altered-number copies 22/22. Field-level OCR accuracy: run eval/ocr_eval.py with an API key before submission and insert the figure.
Usability	Not yet measured - run the 80% task-completion test with 3-5 people before the event if possible.

Known limitations - include these

- The data is synthetic and the abuse patterns were designed by the team, so real-world precision and recall will be lower.
- The lane thresholds were set on this same dataset, so there is a risk of over-fitting. Re-calibrate during the shadow pilot.
- The supplier cross-match assumes invoice-level sales listings or e-invoicing. With only return totals it is weaker.
- A perceptual image hash alone cannot separate re-photographed copies from different invoices on the same supplier template, so duplicates are confirmed by matching fields.
- Low-level input-VAT inflation (under a 10% gap) is the main source of missed abuse (53 claims).
- Response times were measured on one laptop, so they are not a production benchmark.

§10 Costs - add the per-invoice running cost

§10 gives pilot costs in Maloti. Add one row for running costs of invoice reading (at an assumed rate of USD 1 = M 18.5 [assumption]):

Cost item	Indicative (M)	Assumption
Invoice reading - Claude Opus 5.5 (default)	~M 0.37 per invoice	~1.9k input + ~0.5k output tokens; measure with eval/ocr_eval.py
Invoice reading - Claude Haiku 5.5 (budget)	~M 0.009 per invoice	Same token profile; validate accuracy first
Example: 100,000 invoices per year	~M 37,000 (Opus) / ~M 925 (Haiku)	Volume [assumption]

Cost item	Indicative (M)	Assumption
Offline fallback (Tesseract OCR)	M 0	Lower accuracy; manual route remains

Scope note - expatriate employment income

§2.1 lists expatriate employment income and the classification of related-party services among the challenge's risk areas. The prototype has no module for this (the synthetic data includes 40 expat engagements for a future checker). Either add one line saying it is a planned extension, or remove it from the scope list so judges don't expect it in the demo.

Part 3 - Demo script (the brief's three required cases)

Case	Taxpayer & claim	Result	Why
1. Ordinary	Maluti Highlands Knitwear Exporters (Pty) Ltd 2026-10, M 696,724	**Fast-track**, score 0	Claims every month and input VAT is far above output, but M01 (zero-rated exports) explains it. All invoices cross-match. Mitigations: M01.
2. Exception	Senqu Valley FMCG Distributors (Pty) Ltd 2026-10, M 1.3m	**Officer review**, score 0	New business with a huge spike versus its own history. M02 shows it is warehouse equipment, so the only step is one asset document (L02).
3. Human review	Thaba-Bosiu Retail Holdings (Lesotho) (Pty) Ltd 2026-10, M 720,689	**Audit**, score 100	Rules fired: R03, R09, R11, R04, R10, R01, R08, R12. Phantom-supplier invoices, an altered re-photographed copy, low-tax fees, intra-group loan interest, years of losses. Open the group graph.

Suggested 5-minute flow

- Sign in as **auditor1** (password demo) and open the Refund queue: highest risk first, with reason chips.
- Case 1: show the M01 mitigation offsetting the persistence signals, then approve with a reason.
- Case 2: show the M02 explanation and the one-document request, then request documents.
- Case 3: walk the evidence (invoice cross-check, the duplicate image, the group graph), mark one flag "information required", and escalate. Point out that the decision and the reason are in the audit log.
- Scan invoice: pick a sample photo (or upload a PDF) and show auto-filled fields, confidence and the live checks.
- Evidence & evaluation page: precision and recall against the baseline, the limitations, response times.
- Optional: sign in as **senqu** to show the taxpayer portal and the Sesotho toggle.

Before submission - checklist

- Set ANTHROPIC_API_KEY, run *python -m eval.ocr_eval --limit 40* and add the OCR accuracy and cost figures to §8 and §10.
- Economist: confirm the VAT rate (15% used), the turnaround assumptions (3 / 14 / 60 days against ~45 manual) and the USD/LSL rate.
- Native speaker reviews the Sesotho labels.
- Run docker compose once on a machine with enough free memory (it has not been verified yet).
- Re-generate this guide after any change: *python docs/build_update_guide.py*.